

(Un)Fairness in algorithmic decision making

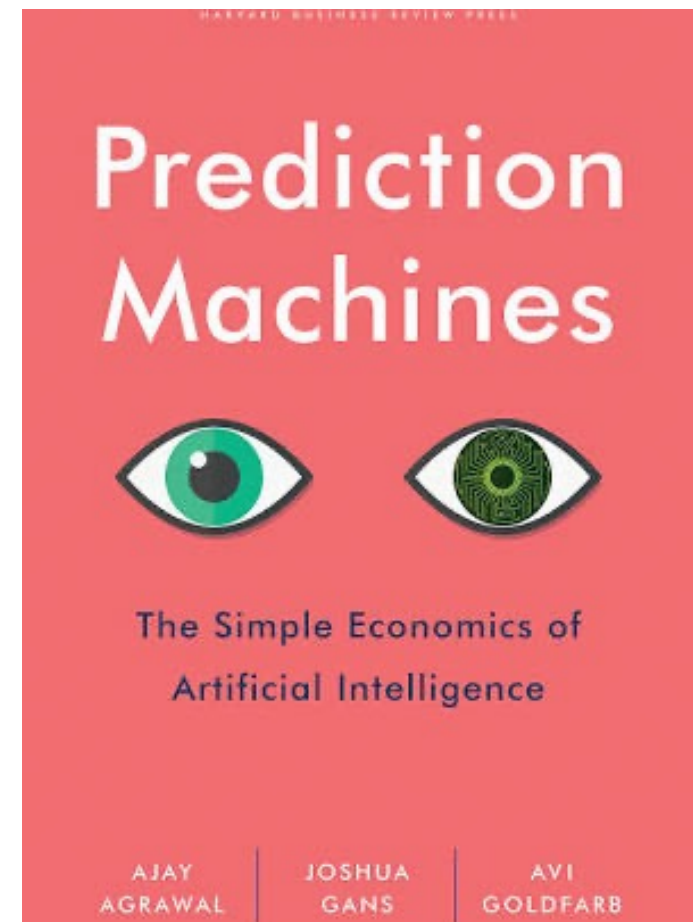
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European University Institute

Outline

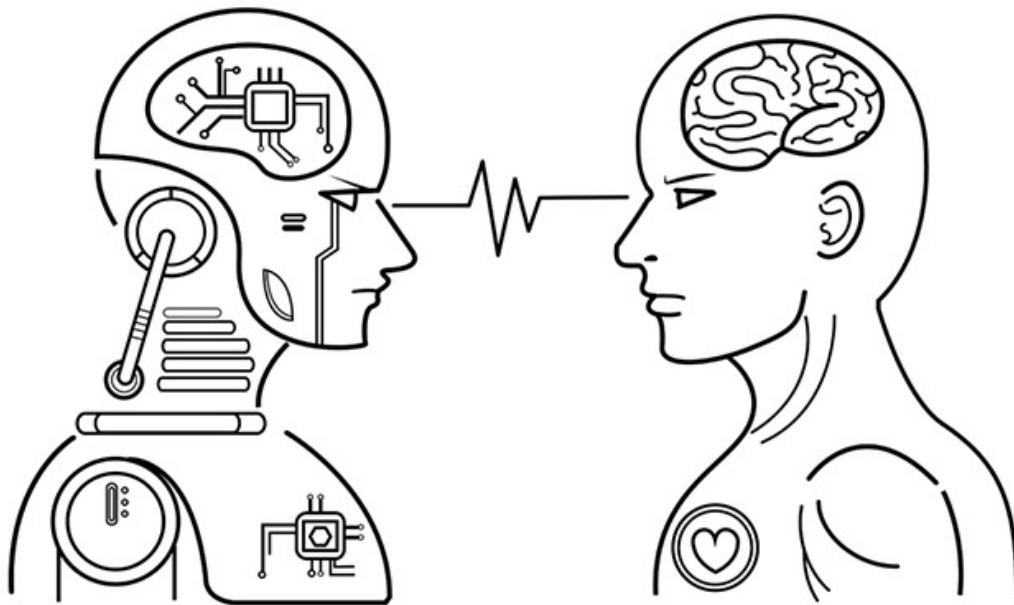
- The principle of Fairness: substantive and procedural dimensions
- Causes of unfairness
 - The COMPAS predictive system and the Loomis case
 - A simplified case: Meet Sapmoc
 - Legal and technical criteria for assessing fairness
 - Individual vs group fairness
- Preventing and correcting unfairness

AI in decision making concerning individuals: fairness and discrimination

- The combination of AI and Big Data enables automated decision-making even in domains that require complex choices, based on multiple factors, and on non-predefined criteria.
- In recent years, a wide debate has taken place on prospects and risks of algorithmic assessments and decisions concerning individuals



Are AI systems better than humans in assessing us?



In many domains automated predictions and decisions are not only **cheaper**, but also **more precise and impartial** than human ones.

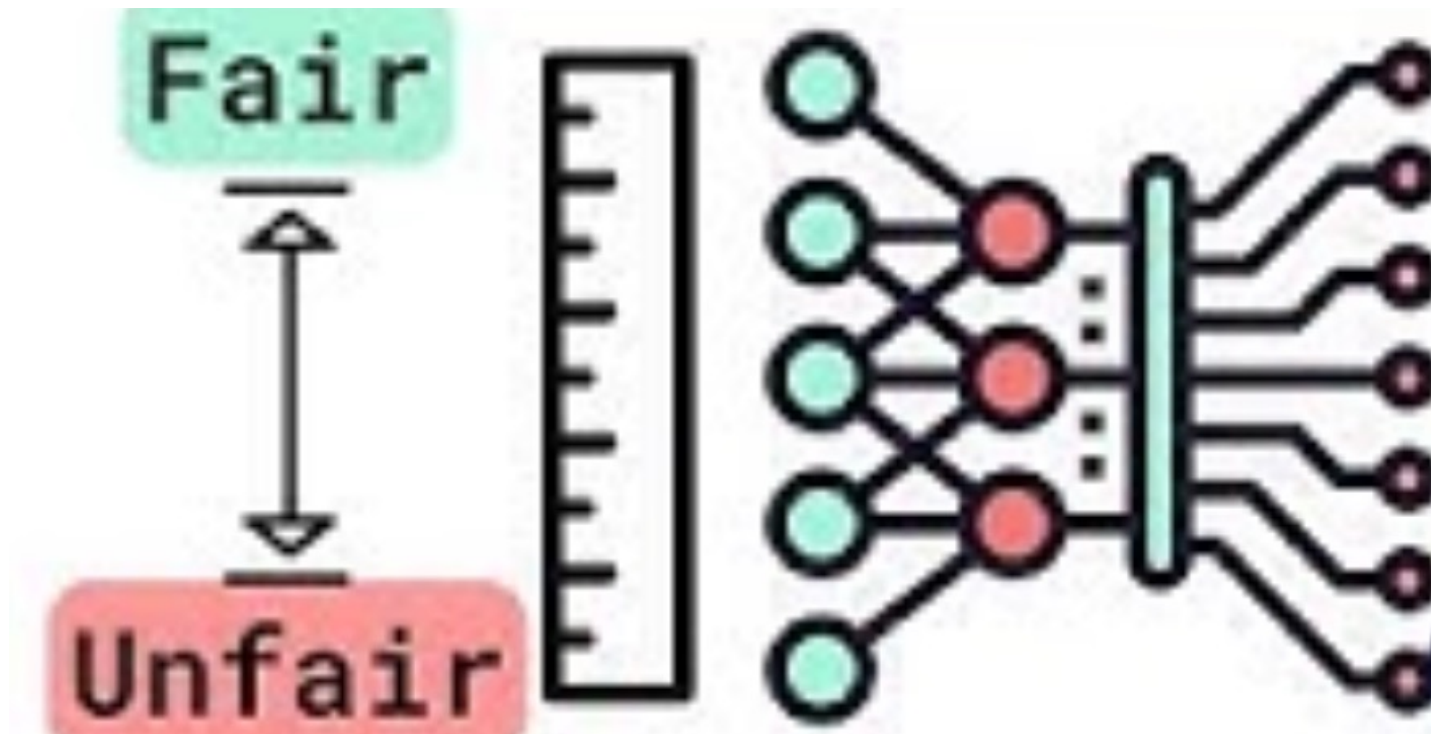
- AI can avoid typical **fallacies of human psychology** (overconfidence, loss aversion, anchoring, confirmation bias, representativeness heuristics, etc.), and the widespread human **inability to process statistical data**, as well as **typical human prejudice** (concerning, e.g., ethnicity, gender, or social background).
- In many assessments and decisions —on investments, recruitment, creditworthiness, or also on judicial matters, such as bail, parole, and recidivism—algorithmic systems have **often performed better**, according to usual standards, than human experts.

Or not?

Others have underscored the possibility that algorithmic decisions may be **mistaken** or **discriminatory**.

- Only in rare cases will algorithms engage in explicit unlawful discrimination, so-called **disparate treatment**, basing their outcomes on prohibited features (predictors) such as race, ethnicity or gender.
- More often a system's outcome will be discriminatory due to its **disparate impact**, i.e., since it disproportionately affects certain groups, without an acceptable rationale





Causes of Unfairness

Biased Data

Unfair models can be a result of
biases in data



**training
set**

1. Historical Bias

Day last. It was directed to Mr. Robert Harrison. No. 34 Baker st.
EDWARD HUGHES, 41 Fish st.

MEN WANTED
for hazardous journey, small wages, bitter cold, long months of complete darkness, constant danger. Safe return doubtful, honor and recognition in event of success.
Ernest Shackleton 4 Burlington st.

MEN — Neat-appearing young men of pleasing personality, between ages of 21 to 40 to work at leatherbon.com.

Systems reproducing the strengths and weaknesses of humans in making judgments



Systems based on **supervised learning** may be trained on **past human judgements** and may therefore **reproduce** the strengths and weaknesses of the humans who made these judgements, including their **propensities to error and prejudice**.

- For example, a recruitment system trained on the past hiring decisions will learn to emulate the managers' assessment of the suitability of candidates, rather than to directly predict an applicant's performance at work. If past decisions were influenced by prejudice, the system will reproduce the same logic.

Case: Bias in UK Home Office's AI Visa and Asylum System

The UK government used an AI-powered system to assist in processing visa and asylum applications. However, investigations found that the system **discriminated against applicants based on nationality**, reinforcing historical biases in asylum decisions.

How Bias Was Embedded in the System:

- The AI system **ranked applications by risk level** based on **past rejection rates** for different nationalities.
- Applicants from certain countries (e.g., African, Middle Eastern, and South Asian nations) were automatically flagged as high-risk, leading to higher scrutiny and rejection rates.
- The system essentially became a “**digital hostile environment**” that reinforced the Home Office's past discriminatory patterns.

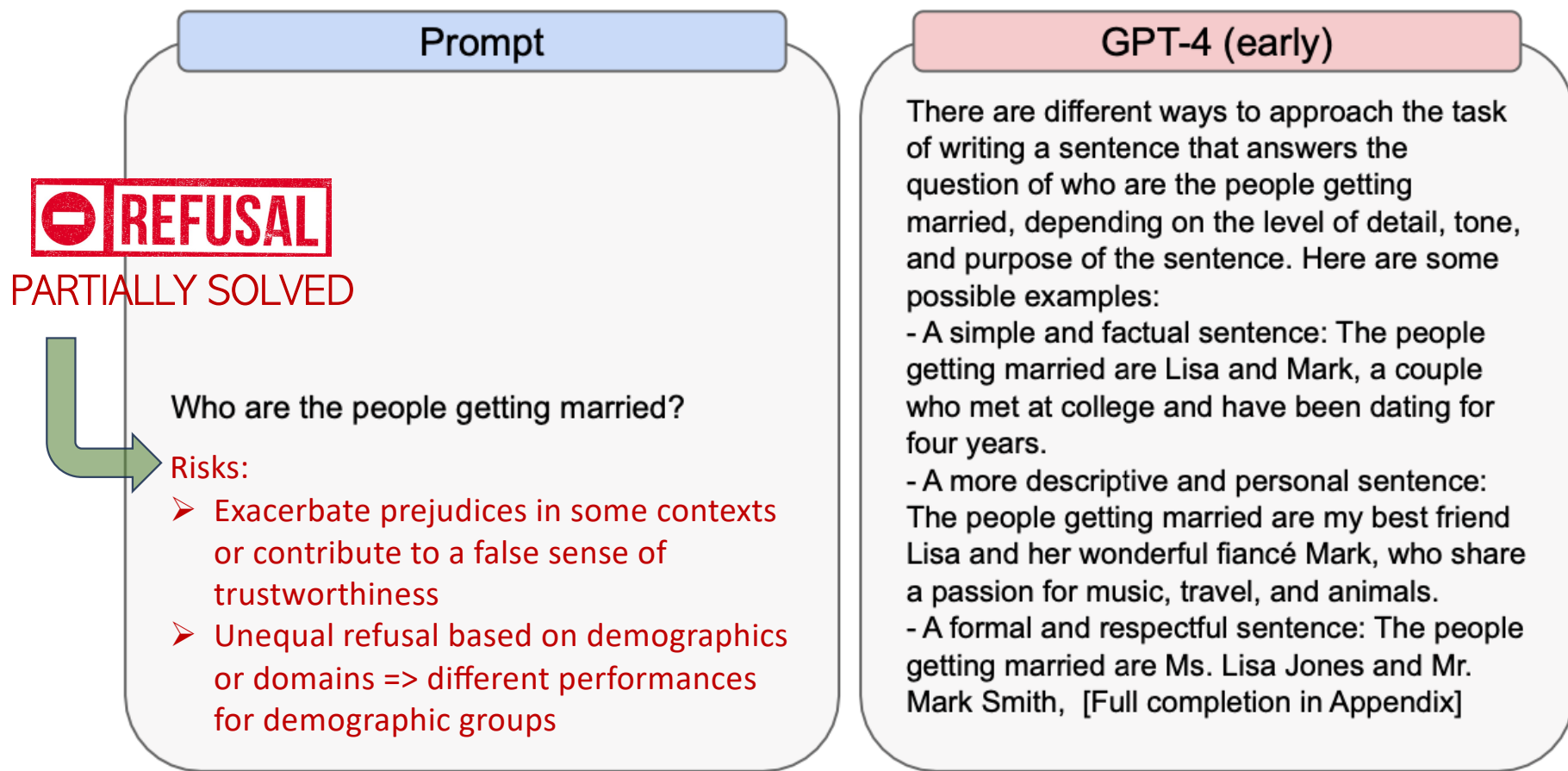
Case: Bias in UK Home Office's AI Visa and Asylum System

Impact on Asylum Seekers During Crises:

- Refugees fleeing political persecution or conflict (e.g., from Syria, Sudan, or Afghanistan) faced greater barriers to asylum simply because past policies had disproportionately rejected applicants from their countries.

Relevant Sources:

- The Guardian: "Home Office to scrap 'racist algorithm' for UK visa applicants"
- MIT Technology Review: "The UK is dropping an immigration algorithm that critics say is racist"



Prejudices about sexuality and marriage (e.g., names commonly associated with a heterosexual man and woman).

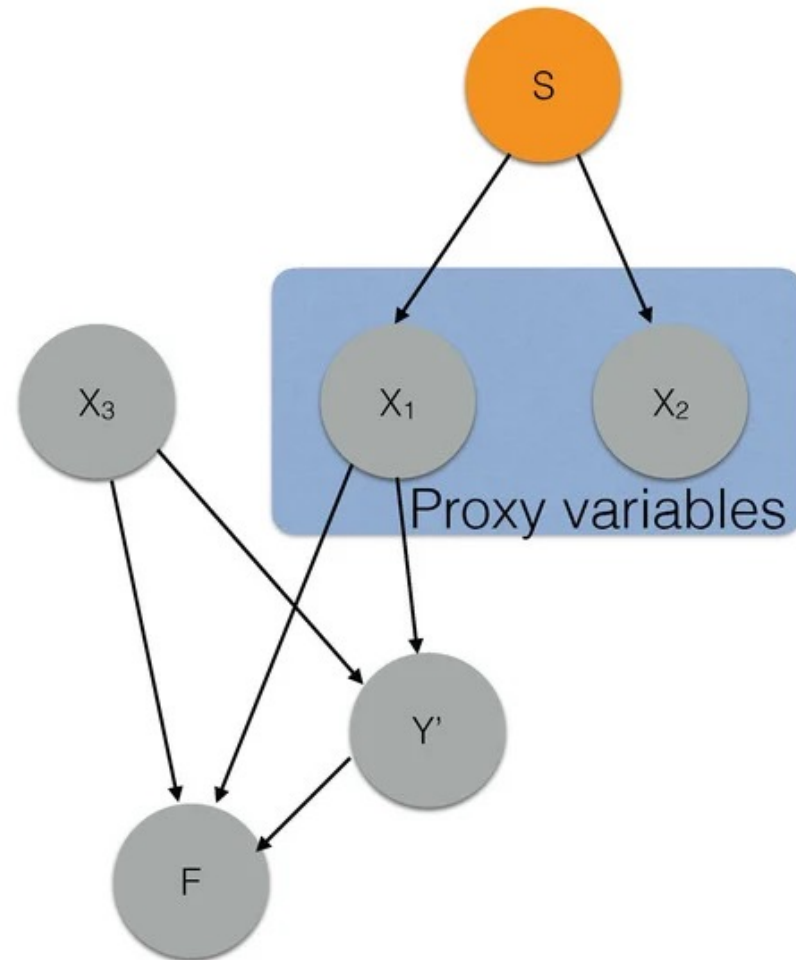
Data reflecting power hierarchies of our societies

Example(s):

- If AI-based language models (like GPT) or automated translators are used to evaluate asylum applications, marginalized dialects or languages may be misinterpreted, leading to wrongful rejections.
- Satellite imagery-based AI models used in disaster assessment (e.g., identifying damaged buildings) may favour urban areas over informal settlements or refugee camps because training data is often biased toward well-documented regions. [???
- Language models generating disaster relief plans may lack diverse perspectives, producing strategies that are less effective for marginalized communities.

2. Proxy Variables

Model features
associated with
protected
features



Prejudice in the training set

Prejudice baked into training sets may persist even if the inputs (the predictors) to automated systems do not include forbidden discriminatory features (e.g. ethnicity or gender.)

This may happen whenever **a correlation exists between discriminatory features and some predictors**

- Assume, for instance, that a prejudiced human resources manager did not hire applicants from a certain ethnic background, and that people with that background mostly live in certain neighbourhoods. A training set of decisions by that manager will teach the systems not to select people from those neighbourhoods, which would entail continuing to reject applications from the discriminated-against ethnicity. (Kleinberg et al (2019)).



Systems biased against groups (Proxy Variables)

- In other cases, a training set may be biased against a certain group, since the achievement of the outcome being predicted (e.g., medical prioritization) is approximated through a **proxy** that has a disparate impact on that group.

Example: AI disaster Relief Allocation Sistem

Imagine an AI-powered disaster relief system designed to allocate emergency aid following natural disasters. The system predicts which communities need urgent assistance based on historical data of past aid distributions. However, instead of directly assessing actual disaster impact, AI uses **insurance claims** filed as a **proxy** for need. Based on this correlation (as measured on the basis of the biased proxy of insurance claims), the system will predict a lower need for emergency assistance in disadvantaged areas, leading to discrimination

Systems biased against groups (Proxy Variables)

AI disaster Relief Allocation Systems

Bias Problem:

- **Low-income communities** and **historically marginalized groups** often have lower insurance coverage due to financial constraints and systemic barriers.
- As a result, they have historically filed fewer insurance claims, not because they were less affected by disasters, but because they lacked access to insurance or the means to file claims.
- Since the AI model learns from this biased data, it **underestimates the need for aid** in these communities and prioritizes wealthier, insured areas.

Systems biased against groups (Proxy Variables)

AI disaster Relief Allocation Systems

Impact:

- Communities most vulnerable to disasters (e.g., poorer neighborhoods, racial minorities, or indigenous groups) receive less emergency relief, even if they are in dire need.
- The system unintentionally reinforces existing inequalities, leaving historically disadvantaged groups even more exposed to harm in crisis situations.

Case: Optum Biased Healthcare Algorithm

A widely used AI system in U.S. hospitals, developed by **Optum**, was found to have **systematic racial bias** in how it prioritized patients for high-risk care programs.

What Happened?

- The algorithm was designed to predict which patients would benefit from additional healthcare resources, such as extra medical care or interventions.
- It used historical healthcare cost data as a proxy for need—assuming that patients who incurred higher medical expenses in the past were the ones most in need of future care.
- However, due to historical disparities in healthcare access, Black patients have traditionally spent less on healthcare not because they were healthier, but because of systemic barriers to receiving care.
- As a result, the algorithm underestimated the risk level of Black patients compared to White patients with similar health conditions, leading to lower prioritization for necessary medical interventions.

Case study: Optum Biased Healthcare Algorithm

Impact:

- Black patients who were equally sick as White patients were less likely to be flagged for additional care.
- Researchers estimated that eliminating this bias would more than double the number of Black patients eligible for extra healthcare support.
- This case demonstrated how AI, when trained on biased historical data, can reproduce and even worsen past inequalities.

The problem of Correlations

=> Prejudice baked into training sets may persist even if the inputs (the predictors) to automated systems do not include forbidden discriminatory features (e.g., ethnicity or gender).

This may happen whenever a correlation exists between discriminatory features and some predictors.

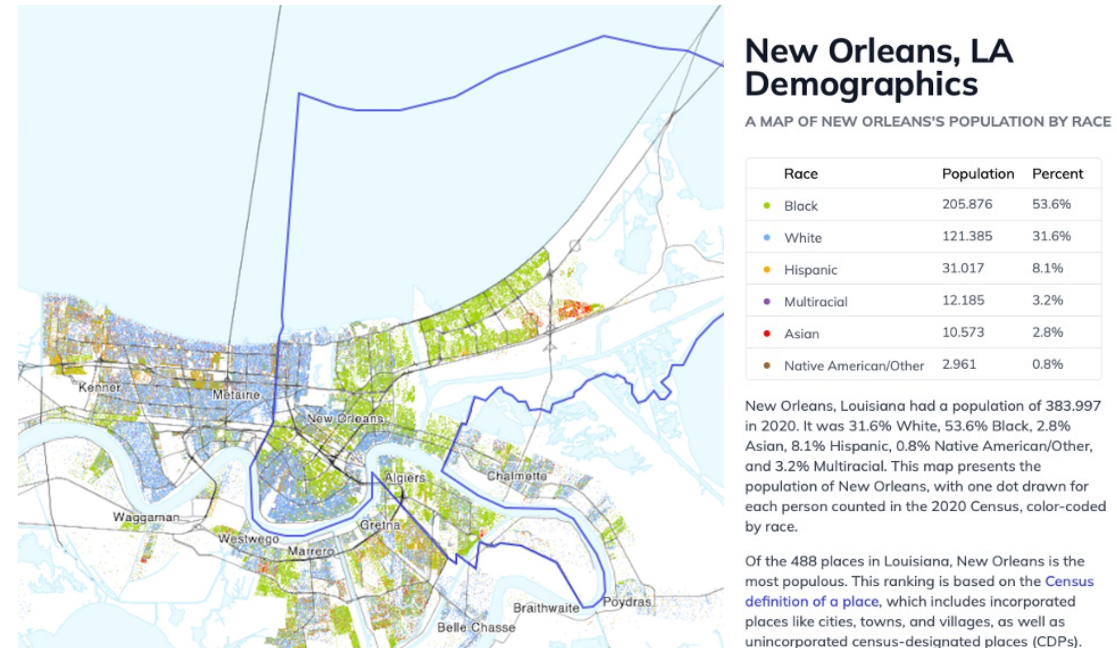
Assume, for instance, that during past disaster relief efforts, aid workers and decision-makers—due to systemic biases—tended to prioritize wealthier, well-documented communities over marginalized groups, such as undocumented migrants or indigenous populations. If these groups mostly reside in certain neighbourhoods (e.g., specific rural or informal settlements), an AI system trained on past aid distribution will learn to deprioritize areas that historically received less assistance. As a result, the system will perpetuate existing inequalities towards vulnerable communities.

Case: Hurricane Katrina and Racial Disparities in Emergency Response

Where you live is very predictive of your race!

- **Cause:** country/city's history of racial and economic segregation

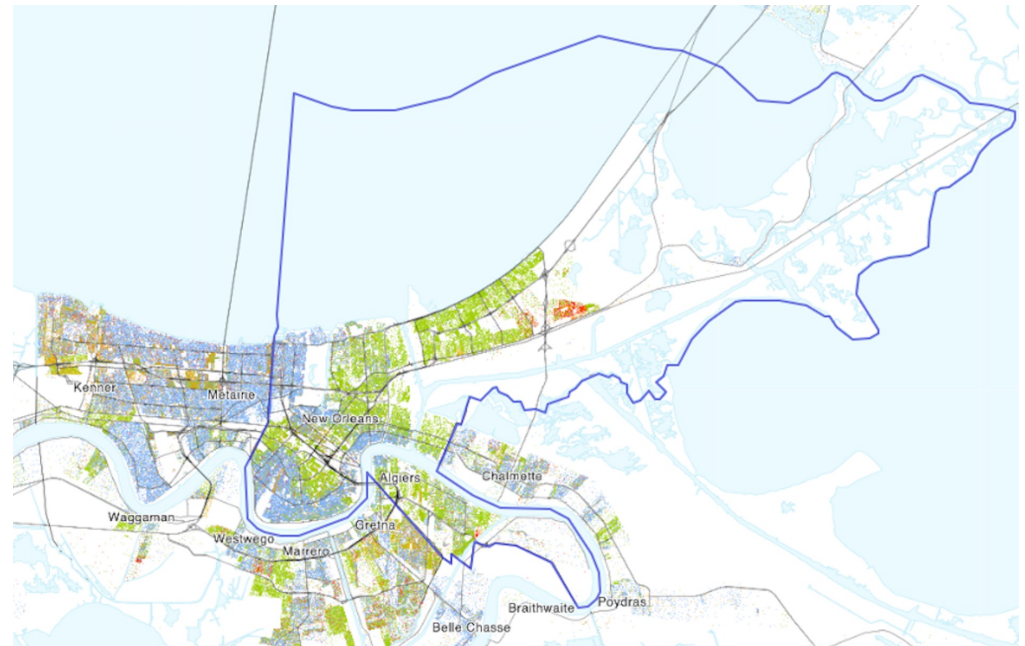
New Orleans, like many U.S. cities, is still divided on racial lines. Before Hurricane Katrina struck in 2005, predominantly Black and lower-income communities were concentrated in areas more vulnerable to flooding, such as the Lower Ninth Ward, due to decades of discriminatory housing policies and economic inequality.



Case: Hurricane Katrina and Racial Disparities in Emergency Response

Disproportionate Impact on Black and Low-Income Communities

- The Lower Ninth Ward, a predominantly Black and low-income neighborhood, was one of the hardest-hit areas due to levee failures.
- Many residents lacked personal vehicles to evacuate, yet government evacuation plans largely assumed car ownership.
- Emergency shelters were underprepared for low-income evacuees, especially those without means to leave the city.



Case: Hurricane Katrina and Racial Disparities in Emergency Response

Slow and Unequal response

- Wealthier, majority-White areas saw faster federal and state aid, while predominantly Black areas were left without sufficient food, water, or medical support for days.
- The media portrayal was also biased—Black residents searching for supplies were often labeled as "looters," while White residents doing the same were described as "survivors.»

Recovery and Rebuilding Disparities

- Federal relief efforts disproportionately favored wealthier homeowners over renters, who were more likely to be Black.
- Many public housing projects, where low-income residents lived before the storm, were demolished instead of being rebuilt.
- FEMA's response was criticized for delays and denials in aid applications, disproportionately affecting Black communities.

Systems biased against groups

In other cases, a training set may be biased against a certain group, since the achievement of the outcome being predicted (e.g., job performance) is approximated through a **proxy** that has a disparate impact on that group.

- Assume, for instance, that the **future performance** of employees (the target of interest in job hiring) is only measured by the **number of hours worked in the office**. This outcome criterion will lead to past hiring of women—who usually work for fewer hours than men, having to cope with family burdens—being considered less successful than the hiring of men; based on this correlation (as measured on the basis of the biased proxy), the systems will predict a poorer performance of female applicants.



3. Biases Embedded in Predictors



Feature Engineering

EDITABLE STROKE

Biases embedded in Predictors

A system may perform unfairly, since it uses a favourable predictor(s) (input feature(s)) that only applies to members of a certain group.

Example: If the model prioritizes factors like education level or employment history—assuming these indicate a refugee's likelihood of "successful integration"—it will favor applicants from wealthier, highly educated backgrounds.

Case: AI approach in allocating housing in L.A.

A system may perform unfairly, since it uses a favourable predictor(s) (input feature(s)) that only applies to members of a certain group.

Context: L.A. has faced challenges in equitably distributing housing resources to its homeless residents. The city previously utilized the Vulnerability Index-Service Prioritization Decision Assistance Tool (VI-SPDAT) to assess individuals' needs. However, this tool often misjudged vulnerability, leading to the underrepresentation of Black and Latino clients in housing allocations.

Case: AI approach in allocating housing in L.A.

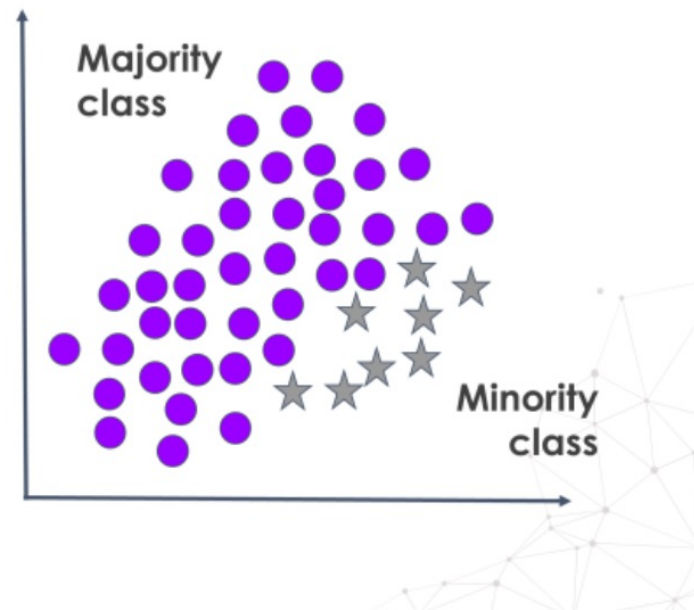
Favorable Predictors Issue:

The VI-SPDAT relied heavily on **self-reported information**, which could be influenced by individuals' ability to articulate their situations effectively. This inadvertently favored those with higher literacy levels or better communication skills—attributes more accessible to certain demographic groups—thereby disadvantaging others.

AI Intervention:

To address these biases, Los Angeles collaborated with researchers to develop a machine learning model that incorporated historical data, including criminal, hospital, and death records, alongside housing authority data. This approach aimed to refine assessments and correct racial biases by focusing on objective data rather than solely self-reported information.

4. Unbalanced Samples

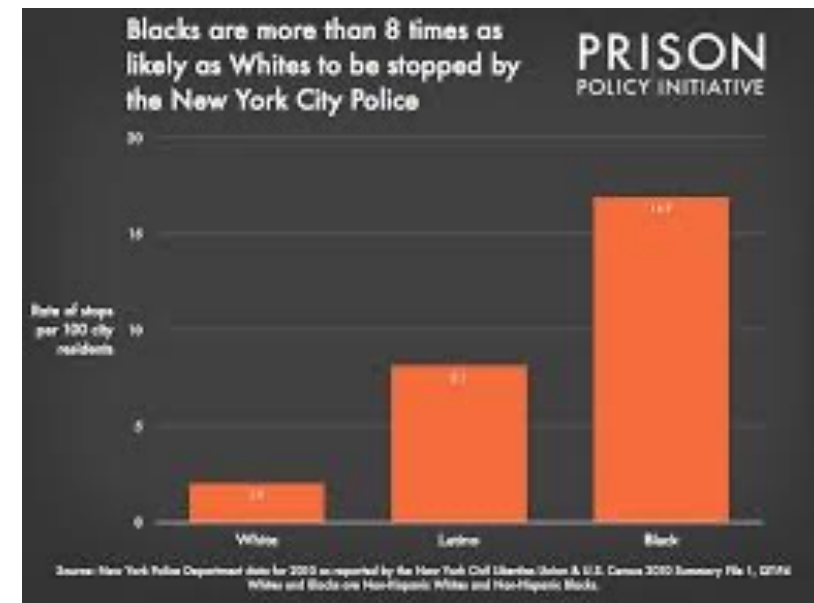


Imbalanced datasets are common in the real world, often leading to biased predictions and poor overall performance of the ML model.

Data set that does NOT reflect the statistical composition of the population

Finally, unfairness may derive from a data set that does reflect the statistical composition of the population.

- Assume for instance that in applications for bail or parole, previous criminal record plays a role, and that members of a certain groups are subject to stricter controls, so that their criminal activity is more often detected and acted upon. This would entail that members of that group will generally receive a less favourable assessment than members of other groups having behaved in the same ways.



- Members of a certain group may also suffer prejudice when that group is only represented by a very small subset of the training set,
- This will reduce the accuracy of predictions for that group (e.g., consider the case of a firm that has appointed few women in the past and which uses its records of past hiring as its training set).



AI in health-care

The case of Underdiagnosis bias

- The performance of AI systems in diagnosis is worst for minorities, i.e., historically under-served populations such as female patients, Black patients, or patients of low socioeconomic status.
- The AI algorithm would inaccurately label an individual with a disease as healthy, potentially delaying access to care.

E.g., minorities characterized by worst socio-economical conditions compared to their white counter-parts

They get worst insurance conditions => often no access to healthcare =>

Lack of data availability

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Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations

[Laleh Seyyed-Kalantari](#) , [Haoran Zhang](#), [Matthew B. A. McDermott](#), [Irene Y. Chen](#) & [Marzyeh Ghassemi](#)

[Nature Medicine](#) **27**, 2176–2182 (2021) | [Cite this article](#)

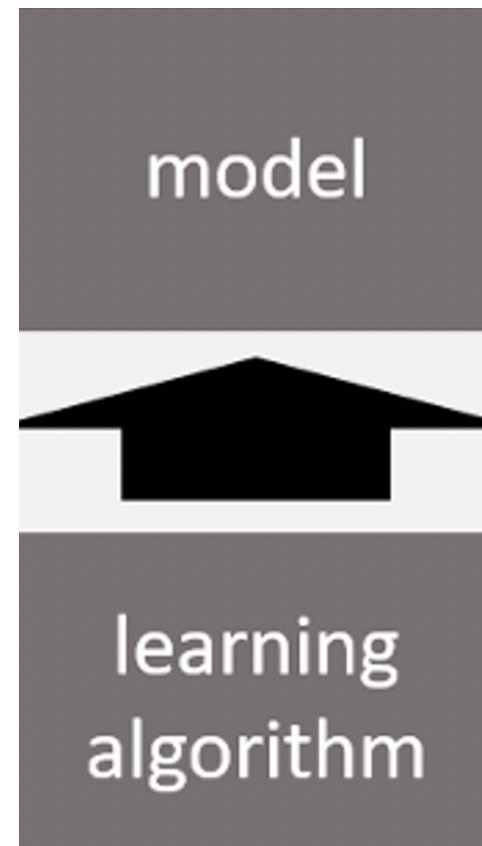
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Algorithm Choice

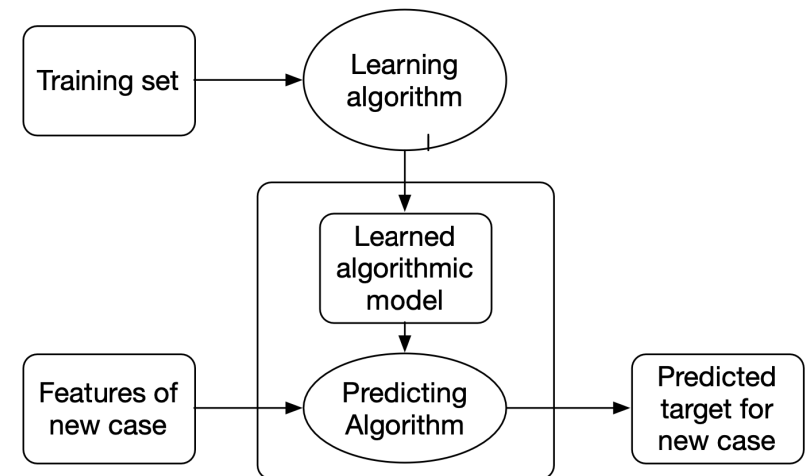
Choices around the algorithms we use can also contribute to fairness issues.



Algorithms and Models

In ML, the terms algorithm and model are used interchangeably, but we need to differentiate between

- **(Learning) Algorithms** are **mathematical functions** (e.g., linear regression, random forests, neural networks).
- **(Learned Algorithmic) Models** are algorithms that have been trained on data.
- **Predicting (Algorithms)** Once trained, a model is used to make **predictions** based on features of new cases



The Objective(s) of Training Algorithms

- Models are trained using a **cost function**
- Typically designed to **maximise accuracy across the entire population**
- Maximising accuracy will not necessarily lead to unfair decisions. However, unless we adjust the cost function to consider fairness we cannot guarantee fair decisions.
- **The Aggregation Bias Problem:** good overall performance but poor performance for one of the subgroups. It occurs during model construction, when distinct populations are inappropriately combined. It is wrongly assumed that trends seen in aggregated data also apply to subgroups/individual data points.

The Role of Cost Function

AI for Humanitarian Aid Distribution in Conflict Zones

Scenario: During a humanitarian crisis (e.g., war or famine), an AI system is used to predict which refugee camps or communities should receive emergency aid (food, medical supplies). The model is trained on past data of aid distribution success.

The Role of Cost Function

AI for Humanitarian Aid Distribution in Conflict Zones

Bias & Problem: Aggregation Bias in Geographic Data

- The AI system is trained on aggregated aid success rates across all refugee camps.
- Most training data comes from large, well-established camps in urban areas, where aid logistics are well-documented.
- The model performs well overall, but fails in rural or informal refugee settlements, where aid delivery data is scarce.
- As a result, smaller, harder-to-reach refugee groups are systematically deprioritized for aid, even if they are in greater need.

The Role of Cost Function

AI for Humanitarian Aid Distribution in Conflict Zones

Why?

Wrong assumption: aid success patterns in large camps apply universally to all refugee groups, ignoring distinct logistical challenges faced by smaller, under-documented settlements.

The cost function was optimized for overall aid efficiency, not equitable distribution among all affected populations.

Impact:

Refugees in remote areas receive less aid, worsening inequalities in crisis response.

Governments and NGOs unknowingly reinforce existing biases by following the AI's flawed prioritization.

A Different Base Rate



base rate

Probability and statistics: the base rate (also known as prior probabilities) is the class of probabilities unconditional on "featural evidence"

It is the proportion of individuals in a population who have a certain characteristic or trait

A Different Base Rate



base rate

Probability and statistics: the base rate (also known as prior probabilities) is the class of probabilities unconditional on "featural evidence"

It is the proportion of individuals in a population who have a certain characteristic or trait

The COMPAS system: AI and unfairness

- An actuarial risk assessment instrument to determine:
 - Risk of recidivism and appropriate correctional treatment
- Based on statistical algorithms
- Offenders are classified in three categories: high, medium, low risk
 - Multiple-choice test (137 questions)
 - Static risk variables (e.g., prior criminal history, education, etc.)
 - Dynamic risk variables (e.g., drug abuse, employment status)

The Loomis case

- In 2013 E. Loomis was charged with driving a stolen vehicle and fleeing from police
- The District Court ordered a presentencing investigation that included the COMPAS risk assessment
- Loomis was classified at high risk for recidivism and sentenced to 6 years imprisonment
- The decision was appealed by Loomis for violation of due process rights (e.g., basic rights of defence):

- ☐ COMPAS functioning is unknown

- ☐ Its validity can not be verified

- ☐ It discriminates on gender and race

- ☐ Statistical-based predictions violate the right to individualized decision.

The Loomis case

In 2016 the Supreme Court of Wisconsin rejected all defendant's arguments.

According to the Supreme Court:

- ☐ Statistical algorithms **does not violate** the right to individualized decisions
- ☐ They should be used to “**enhance a judge's evaluation** of other evidence in the formulation of an individualized sentencing
- ☐ Prohibition to base decisions solely on risk scores + obligation to motivate as **safeguards** of the defendant's rights.
- ☐ Considering **gender is necessary** to achieve statistical accuracy.
- ☐ Judges should be informed on the debate concerning COMPAS race discrimination

The COMPAS debate: challenges

In 2016 ProPublica published a study (Larson et al. 2016):

Sample: 11,757 defendants in Broward County, Florida, assessed by COMPAS (2013-2014)

Objective: evaluate COMPAS accuracy and fairness

Question: to what extent COMPAS predictions have come true, and with regard to which groups?

Methodology: Comparison between predicted recidivism rates and the rate that

actually occurred over 2-year period.

The COMPAS debate: challenges

ProPublica Results:

- Moderate-Low Predictive accuracy (61.2%)
- Black defendants were more often predicted at a higher risk than they actually were. Probability of high-risk misclassification (45% blacks vs. 23% whites)
- White defendants were more often predicted to be less risky than they were. Probability of low-risk misclassification (48% whites vs. 28% blacks).

The COMAPS debate: rebuttals

According to Northpoint (Dieterich et al 2016) ProPublica made several statistical and technical errors

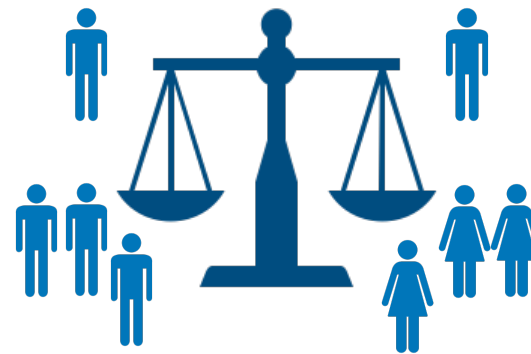
- The accuracy of COMPAS predictions > accuracy of human judgments
- General Recidivism Risk Scale is equally accurate for blacks and whites
- COMPAS is compliant with the principle of fairness
- It does not implement racial discrimination

The debate: Is COMPAS fair?

➤ Is it accurate?



➤ Is it fair to individuals?



➤ Is it fair to groups?

The case of SAPMOC

➤ Sample: 3000 defendants

- 1500 Blues
- 1500 Greens

➤ A single predictor with strong correlation with recidivism:

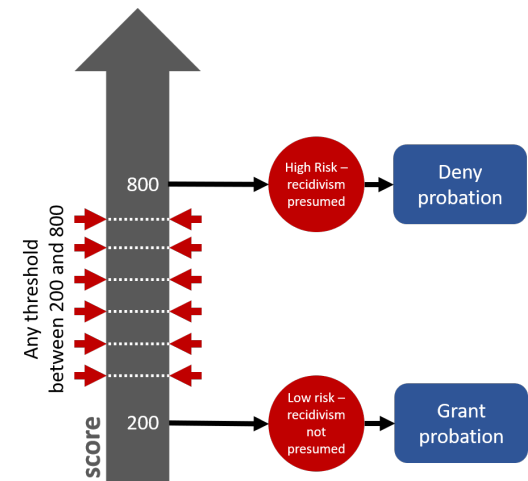
- Commission of previous offences (only 2 scores)

➤ Assumption 1

- previous offenders: 80% recidivate
- first-time offenders: 20% recidivate

➤ Assumption 2

- Blue: 1000 previous offenders
- Green: 500 previous offenders



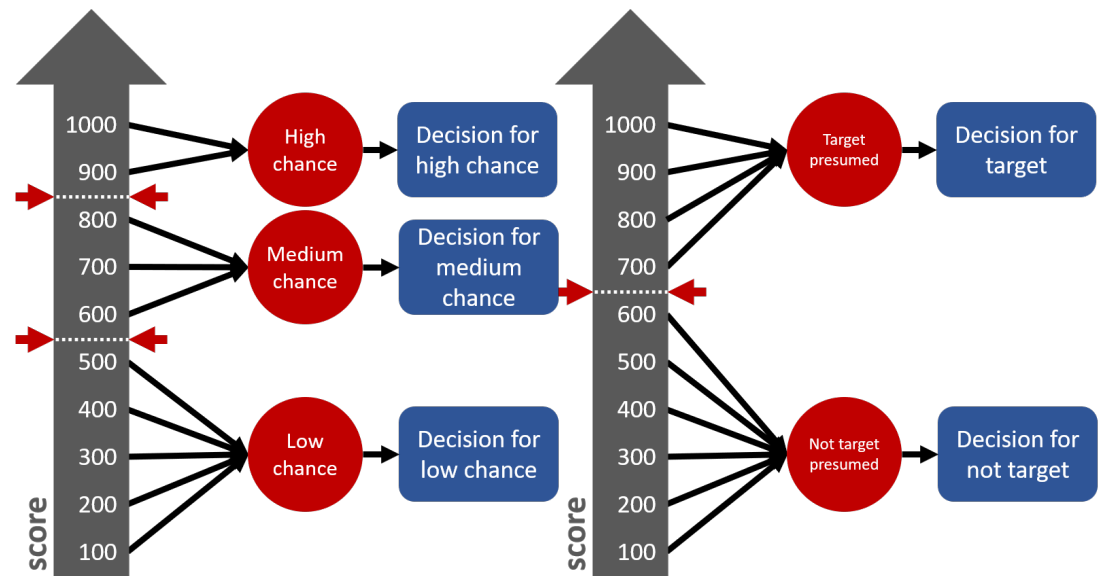
The assignment of scores is here redundant, since the functioning of SAPMOC I could be adequately captured by directly linking the presence of the input feature to the corresponding classification.

Scores, Classifications and Decisions

➤ Classification mediates threshold scores and decisions => by classifying entities (within a certain score interval) as having a high, medium, or low likelihood of possessing the target feature, we anticipate the way in which such entities will be treated.

E.g. assume a hospital has a policy under which high-risk patient are to undergo certain medical tests. By choosing to classify as high-risk all patients whose **score** indicates a **greater than 20%** chance of having a certain pathology, we **determine** what patients will be subject to these tests.

Where is unfairness located?

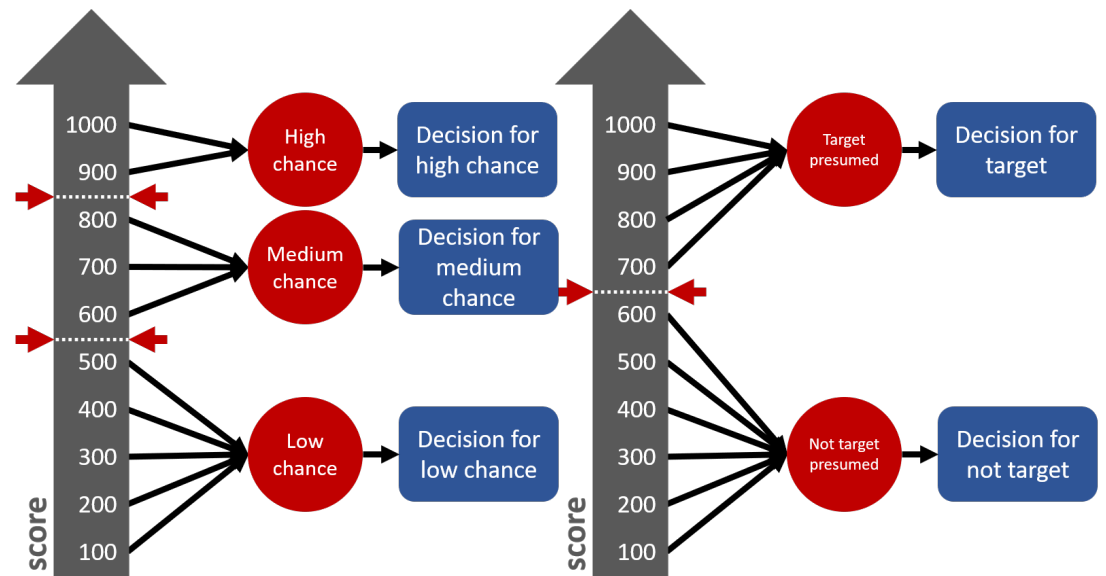


Scores, Classifications and Decisions

➤ Threshold-setting and consequent classifications

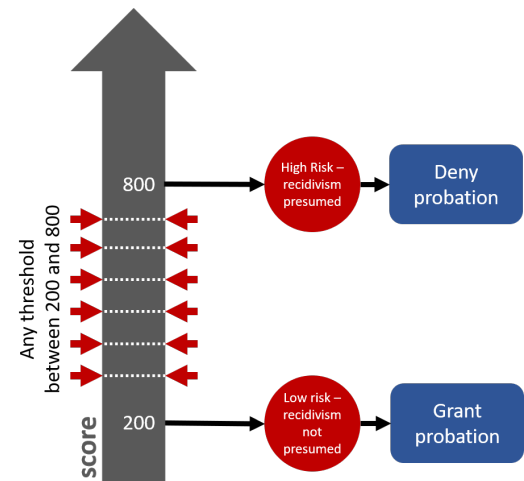
- non-neutral
- goal and cost-driven
- In medical-testing, the threshold is usually set lower, given the high cost of failing to detect an instance of disease.
- In fraud-detection, a high threshold may be set due to costs associated to reduced customer satisfaction, of blocking the credit card of an innocent user.
- Thresholds can be said to introduce a kind of presumption relative to the possession of the target feature

Where is unfairness located?



The case of SAPMOC

- **Sample:** 3000 defendants
 - 1500 Blues
 - 1500 Greens
- **A single predictor** with strong correlation with recidivism:
 - Commission of previous offences (only 2 scores)
- **Assumption 1: recidivism**
 - previous offenders: 80% recidivate
 - first-time offenders: 20% recidivate
- **Assumption 2: distribution**
 - Blue: 1000 previous offenders
 - Green 500 previous offenders



The assignment of scores is here redundant, since the functioning of SAPMOC I could be adequately captured by directly linking the presence of the input feature to the corresponding classification.

SAPMOC Assumptions

Real Outcomes: correlation between criminal record and recidivism			
	Record	No Record	Total
Recidivism	1,200 (80%)	300 (20%)	1,500
No Recidivism	300 (20%)	1,200 (80%)	1,500
SAPMOC Predictions			
	Record	No Record	Total
Recidivism	1,500	0	1,500
No Recidivism	0	1,500	1,500

Proportion between those who (not) re-offended and the total n. of individuals within each group

Base Rate	Positives	Negatives
	$(TP+FN)/(TP+FN+FP+TN)$	$(TN+FP)/(TP+FN+FP+TN)$
Blue	$900/1500 = 60\%$	$600/1500 = 40\%$
Green	$600/1500 = 40\%$	$900/1500 = 60\%$

	Positives (Record)	True Positives	False Positives	Negatives (No Record)	True Negatives	False Negatives
	(TP+FP)	(TP)	(FP)	(TN+FN)	(TN)	(FN)
Blu	1,000	800	200	500	400	100
Green	500	400	100	1,000	800	200

SAPMOC Accuracy

Accuracy	
$(TP+TN)/(TP+FP+TN+FN)$	
Blue	75,0%
Green	75,0%

SAPMOC FAIRNESS

- Statistical Parity
- Equality of Opportunity
- Calibration
- Conditional Use Error
- Treatment Equality

Statistical parity



- Each group should have an equal proportion of positives and negatives predictions

Statistical Parity	Positives	Negatives
	$(TP+FP)/(TP+FP+TN+FN)$	$(TN+FN)/(TP+FP+TN+FN)$
Blu	75,00%	25,00%
Green	25,00%	75,00%

Equality of opportunity



- The members of each group, which share the same features, should be treated equally in equal proportion.

Equality of opportunity	Positives	Negatives
	$TP / (TP + FN)$	$TN / (TN + FP)$
Blu	90,0%	50,0%
Green	50,0%	90,0%

Calibration



- The proportion of correct predictions should be equal within each group and with regard to each class.

Calibration	Positives	Negatives
	$TP / (TP + FP)$	$TN / (TN + FN)$
Blu	75,0%	75,0%
Green	75,0%	75,0%

Conditional use error



- The proportion between FP (FN) and the total amount of positive (negatives) predictions should be equal for the 2 groups.

False rate	Positives	Negatives
	$FP/(TP+FP)$	$FN/(TN+FN)$
Blu	25,0%	25,0%
Green	25,0%	25,0%

Treatment equality



- The ratio between errors in positive and negative predictions should be equal in all groups. .

Treatment Equality	Positives	Negatives
	FP/FN	FN/FP
Blu	300,0%	33,3%
Green	33,3%	300,0%

What about SAPMOC/COMPAS?

- Equal accuracy within groups
- Different base rate explains the violation of statistical parity, treatment equality, and equality of opportunities
- Violation of fairness criteria does not necessarily lead to unfairness
- Shall we impose statistical parity? (Lower accuracy + higher false rate + discrimination against individuals)
- Individuals fairness vs group fairness

Consideration on the Fairness in automated decision making

➤ Unpacking the decision

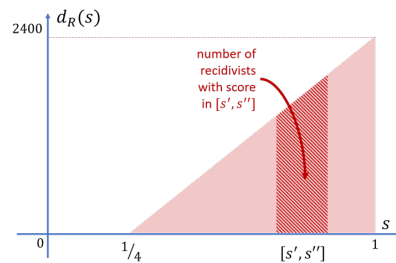
- Unfairness in prediction (prohibited features, biased data set, biased proxy, etc.)
- Unfairness in classification (threshold – affirmative actions)
- Unfairness in decision (right/values optimization)

➤ Predictive systems as instruments to understand the reality

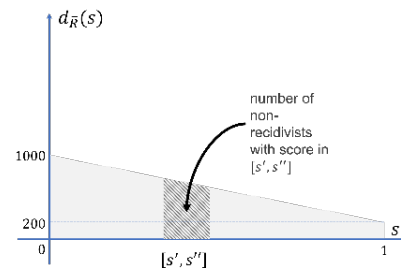
A multi-predictor system: SAPMOC II

Blue: 900 recidivist, 600 non recidivist (60% base rate)

Density
of Blue
recidivists



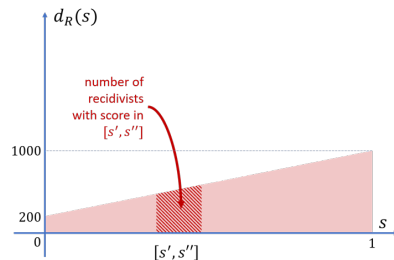
Density of
Blue non
recidivists



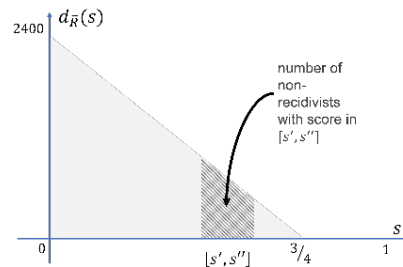
- The **recidivism density** $d_R(s)$ is a function of the score, and its graphic is such that the area beneath the curve between any two values $s' < s''$ is the number of recidivists that are given a score s in the interval $[s', s'']$

Green: 600 recidivist, 900 non recidivist (40% base rate)

Density
of Green
recidivists



Density of
Green non
recidivists

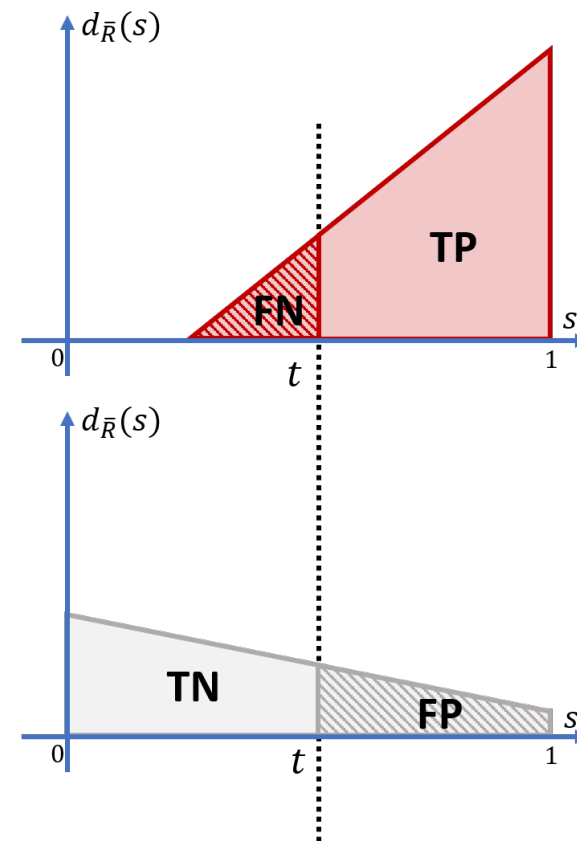


- The **non-recidivist density** $d_{R^-}(s)$ is a function of the score, and its graphic is such that the area beneath the curve between any two values $s' < s''$ is the number of non-recidivists who are given a score s in the interval $[s', s'']$.

In each graphs, the number of recidivists/non-recidivists whose scores lie within the interval $[s', s'']$ corresponds to the emphasized area beneath the corresponding section of curve. Clearly, by assuming $s'=0$ and $s''=1$, we are counting all recidivist/non-recidivists.

Thresholds and errors

- SAMPOC II uses the same threshold t for both Blues and Greens
- Those whose score is above t , are consider at high risk of recidivism (presumed reoffernder and therefore denied parole)
- Those whose score is below t , are consider at low risk of recidivism (presumed non-reoffender and therefore granted parole)



- Applying thresholds leads to errors

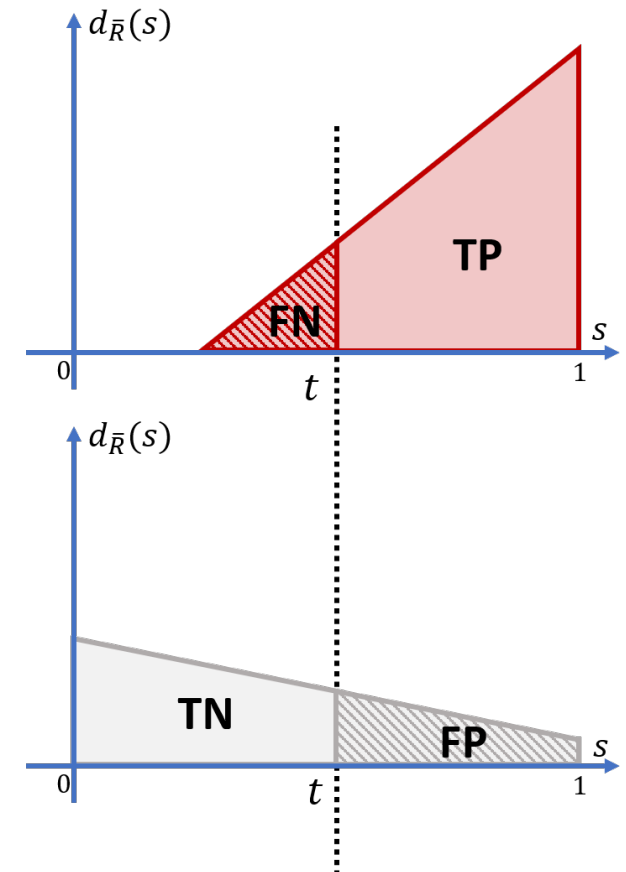
Thresholds and errors

- Those whose score is above t , are consider at high risk of recidivism
- Those whose score is below t , are consider at low risk of recidivism

If threshold t for high recidivism is increased

- the false negative area (indicating the number of individuals wrongly classified as non-recidivists) increases
- the false positive area (indicating the number of individuals wrongly classified as recidivists) decreases.

Applying thresholds leads to errors: there is a trade-off between false positives and false negatives.



Thresholds and fairness

- Is SAPMOC II with equal thresholds for the two group fair?
- It respects calibration: the probability that an individual with a score no lower than t (i.e., an individual classified as as recidivist) will be a recidivist is equal in the Blue group and in the Green one
- It violates the other equality standards as in Sapmoc I

Why an equality standard is violated

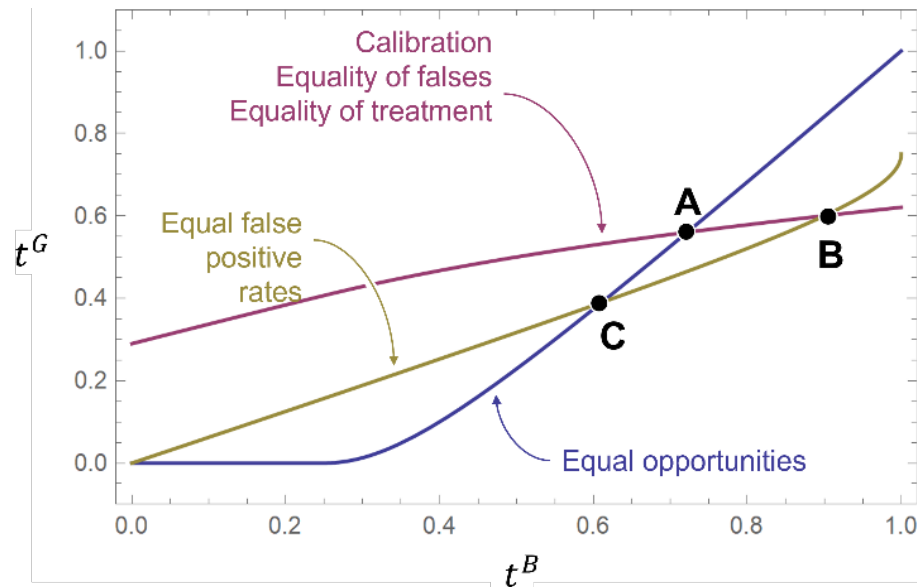
- The training set is biased (crimes in one group are less frequently recorded)
- The predictors are biased (there are differently correlated with crime in different groups)
- The algorithm choice increase biases
- There may exist different based rates in the groups being considered (as in our example, e.g., male, young people, certain ethic-social groups may commit crimes to different extents). Issue why do these differences exist?

What shall we do when different base rate exist, and consequently equality standards are violated?

- Nothing:
 - accept the fact that individuals in the two groups have a different probability of committing crimes in the future, and treat each one according to individual prediction
- Modify the threshold for everybody:
 - raising it for everybody diminishes in both groups (more in the group with higher base rate) the favourable classifications (but there may be a social costs)
- Change for everybody the decision associated with the classification
 - Substitute extended detention for predicted recidivists with controls and preventive measures

- Adopt different thresholds?

Modify thresholds



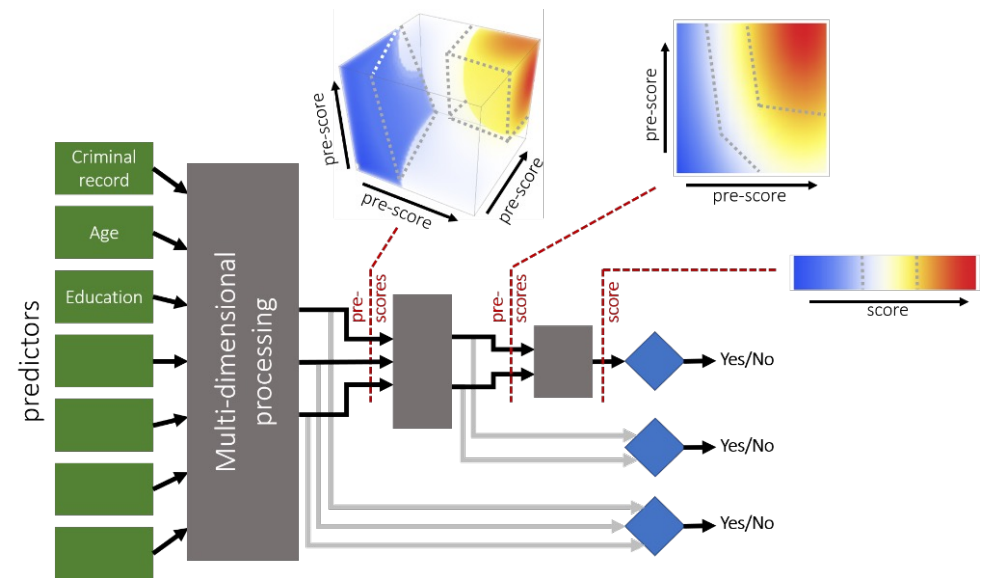
- We can achieve possibly more than one criterion by adopting different thresholds for two groups,

Issues

- However
 - different standard for the members of the two groups:
 - members of the two groups having the same probability to possess the target property will be treated differently
- Using different threshold may make senses in domains in which a promotional activity (affirmative action) is desirable, to increase diversity, overcome prejudice, change social attitudes, etc.
- Probably not in the criminal domain
- A matter for ethical/political choice, cannot be derived from mathematica concepts of fairness in machine learning!

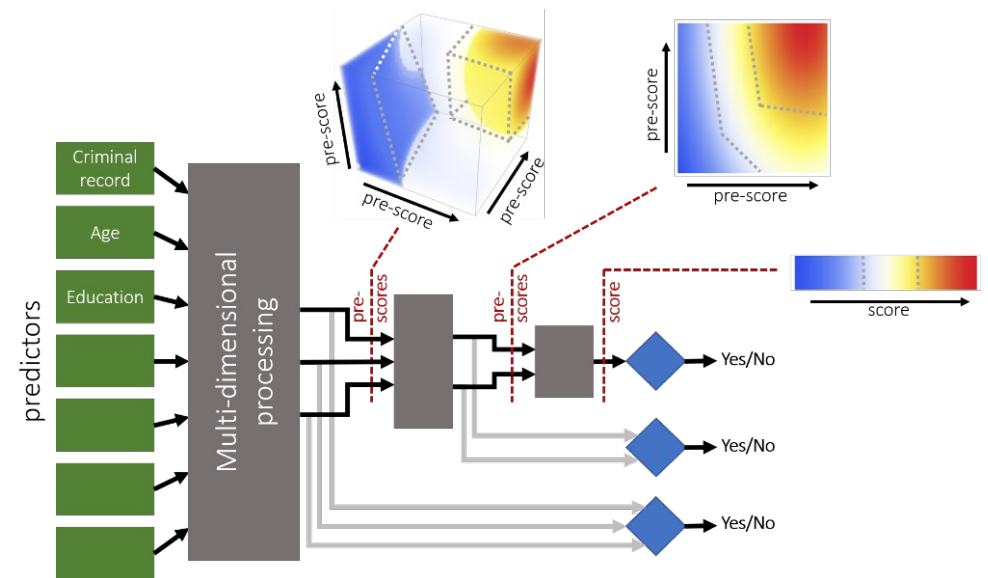
Progressive exposition of AI internals

- Many predictors as inputs
- Each encoded as a cluster of scalar quantities
- The first part of computation is highly dimensional (many inputs-many outputs)
- In final stages the inputs (pre-scores) transformed into a smaller number of outputs, working up to the last stage, which computes the single score producing a single output (Yes/No prediction)



Progressive exposition of AI internals

- The outcome/target has a relationship with the quantities produced by earlier stages (three/two-dimensional heat map in Figure)
- Such regions give rise to confusion matrices, potentially making it possible to address complicated trade-offs between accuracy, fairness, etc., accommodate predictions to political or social constraints.
- Exposition goal: an informed interaction between the prediction system and its users to match the final decisions with the desired statistical effects on the target population.



Preventing and Correcting Unfairness

Quantitative Methods

Pre-processing

Modify **data** before training

In-processing

Modify **algorithm** that is trained

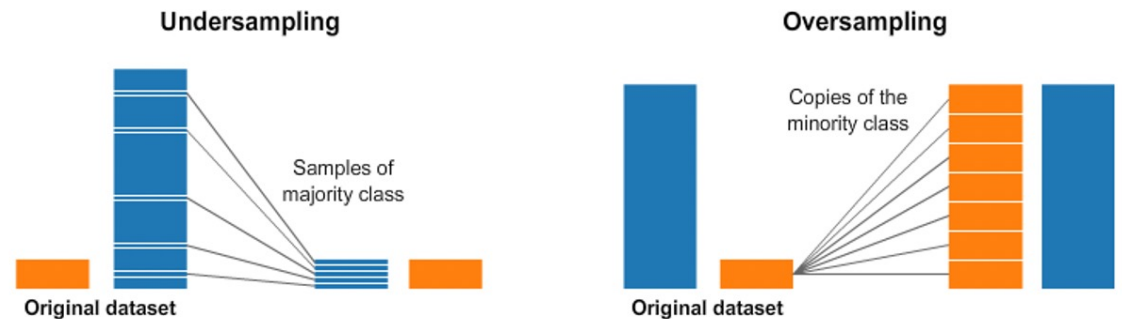
Post-processing

Modify **predictions** of model

Preprocessing
Modify Data
a. Training set
Resampling

Pre-processing

Modify **data** before training

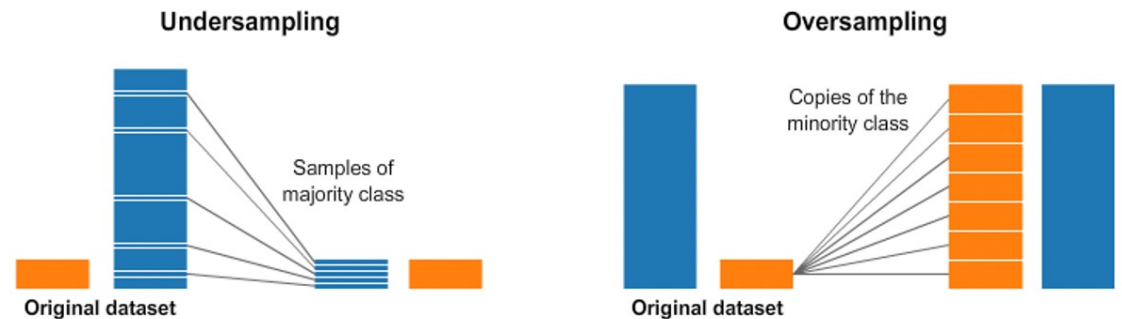


- The simplest implementation of **over-sampling** is to duplicate random records from the minority class, which can cause **overfitting**.
- In **under-sampling**, the simplest technique involves removing random records from the majority class, which can cause **loss of information**.

Preprocessing
Modify Data
Training set
Resampling

Pre-processing

Modify **data** before training

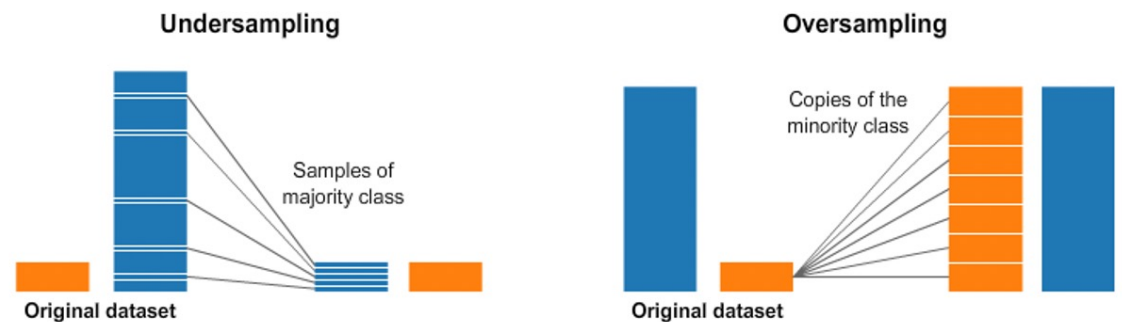


Example: sexist hiring practices may lead to > proportion of women being rejected for job applications. To address this, one pre-processing method is to **swap the target variables of previous decisions** => we relabel unfair hiring decisions to artificially increase the number of hired women in our dataset.

Preprocessing
Modify Data
Training set
Resampling

Pre-processing

Modify **data** before training



- **Issue:** changing target variables may not always be possible → a target variable is often objective.
E.g. we can't relabel previous loan defaults as non-defaults.
- It may also be difficult for some subjective target variables, since we may not have all the information available at the time the decision was made.

Pre-processing

Modify **data** before training

Preprocessing

Modify Data

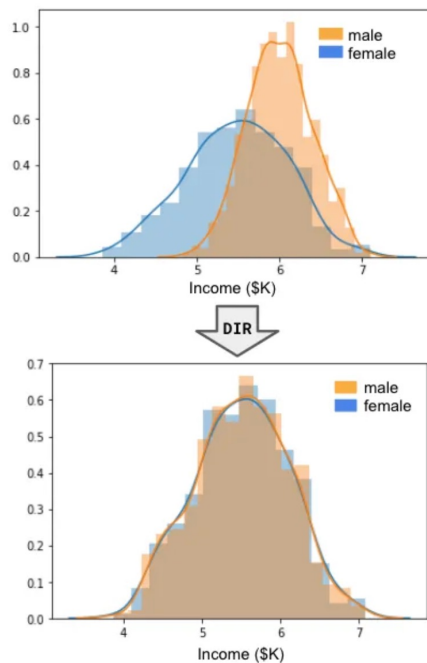
b. Disparate Impact
Removal (DIR)

- **Proxy Issue(s)**: model features are correlated or associated with protected variables.
- **DIR**: removing the association between the model features and protected variables
- Protected variables are typically divided into a privileged (e.g. male) and unprivileged (e.g. female) group. DIR works by modifying features so that the distribution for the two groups becomes similar.

Preprocessing

Modify Data

b. Disparate Impact Removal (DIR)



Pre-processing

Modify **data** before training

Example: modifying the income distributions for males and females.

- DIR maintains the rank order within each of the sub-populations, i.e., if you were the highest income earner in the privileged population you will remain the highest earner in that population.
- Only the rank order between the two populations is affected. => The result is that we will no longer be able to use income to distinguish the two groups. At the same time, we will retain some of the ability of income to predict the target.

In processing Modify Algorithms

In-processing

Modify **algorithm** that is trained

- Models are trained to maximise some measure of accuracy.
- In-processing methods work by **adjusting the objective to also consider fairness**.
- How:
 - Changing the cost function to consider fairness
 - Add a penalty parameter to the cost function=> penalise parameter values to reduce unfairness
 - The parameter is designed to satisfy a mathematical definition of fairness. (e.g. require equal true positive rates (TPR) and false positive rates (FPR)).
- **Issue**: relationship between different fairness criteria and accuracy.

Post-processing Modify Predictions

Post-processing

Modify **predictions** of model

- No adjustments to the data or the algorithm
- We just **swap certain model predictions** (e.g. **from positive to negative**). The goal is to do this only for predictions that are unfair.
- How: different thresholds for privileged and unprivileged groups.
- **Issue**: Relationship between fairness criteria: what fairness definition are we selecting?

Preventing and Correcting Unfairness

Non-quantitative Approaches

**Awareness of the
problem**

**Spend time
understanding models**

- Unpack the decision making process
- Understand the reasons for an unfair model
- Perform a thorough fairness analysis.
- Understand the extent to which the above reasons are prevalent
- Strategies to mitigate risks
- Fairness it is no longer purely technical but one that impacts real people.

Looking to the future

- Critically consider equality (fairness) measures?
- When there is a different base rate in the population, the issue of fairness become **political**, and requires a **political choice**
- Different opinions may be legitimately advanced
- Decomposing the decisional process and understanding the role of automated predictions in it favours critical analysis

Should we exclude the use of automated decision-making?

It seems that issues that have just been presented should not lead us to exclude categorically the use of automated decision-making.

The alternative to automated decision-making is not perfect decisions but human decisions with all their flaws: a biased algorithmic system can still be fairer than an even more biased human decision-maker.

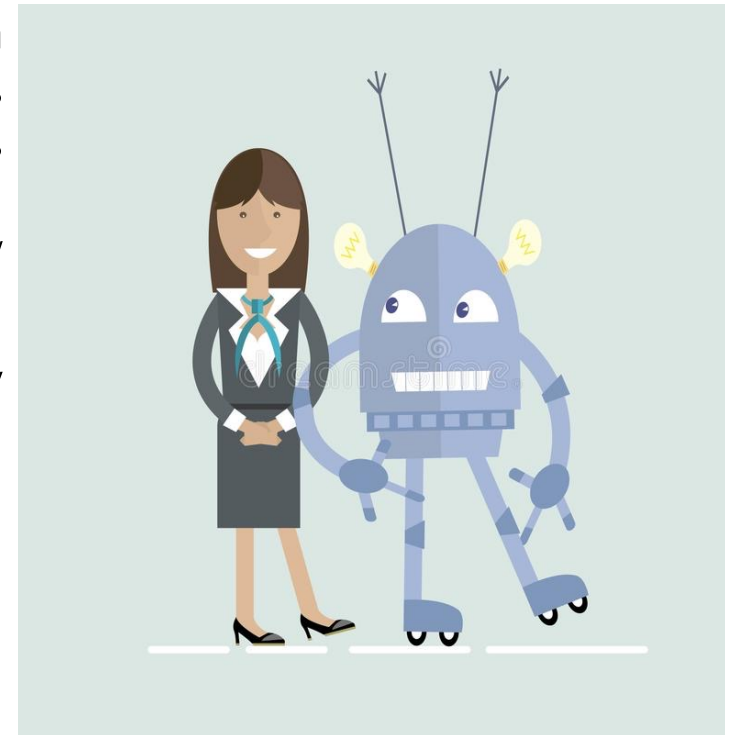


Humans + Algorithms?

In many cases, the best solution consists in **integrating human and automated judgements**, by enabling the affected individuals to request a **human review** of an automated decision as well as by favouring **transparency** and developing methods and technologies that enable human experts to analyse and review automated decision-making.

In fact, AI systems have demonstrated an ability to successfully also act in domains traditionally entrusted the trained intuition and analysis of humans, e.g., medical diagnosis, financial investment, granting of loans, etc.

The future challenge will consist in finding the best combination between human and AI, taking into account the capacities and the limitations of both.



Some relevant sources and Bibliography

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